Unit 2 Ext. Problems

Greyson Knippel
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Problem 1: Fourier Analysis

write-up

functions can be treated like vectors. the inner product is $\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,dx$ (like a dot product but with an integral). if it's 0, the functions are "orthogonal." a fourier series writes a periodic function in the basis $\{1, \cos x, \sin x, \cos 2x, \dots\}$, and the coefficients are its coordinates. we didn't cover this in class, so i loosely learned it from:

- "But what is a Fourier series? From heat flow to drawing with circles" (youtube, 3Blue1Brown) `https://www.youtube.com/watch?v=r6sGWTCMz2k`
- more ai tools since i was pretty confused on this one.

work

A.

i.

$$\langle 1, \cos x\rangle = \int_{-\pi}^{\pi} 1 \cdot \cos x\,dx = \sin x\Big|_{-\pi}^{\pi} = \sin(\pi) - \sin(-\pi) = 0 - 0 = \boxed{0}$$

ii.

$$\langle 1, \sin x\rangle = \int_{-\pi}^{\pi} 1 \cdot \sin x\,dx = -\cos x\Big|_{-\pi}^{\pi} = -\cos(\pi) + \cos(-\pi) = -(-1) + (-1) = \boxed{0}$$

iii. use the identity $\sin x \cos x = \frac{1}{2}\sin 2x$:

$$\langle \sin x, \cos x\rangle = \int_{-\pi}^{\pi} \sin x \cos x\,dx = \int_{-\pi}^{\pi} \tfrac{1}{2}\sin 2x\,dx = -\tfrac{1}{4}\cos 2x\Big|_{-\pi}^{\pi}$$

$$= -\tfrac{1}{4}\cos(2\pi) + \tfrac{1}{4}\cos(-2\pi) = -\tfrac{1}{4}(1) + \tfrac{1}{4}(1) = \boxed{0}$$

all 0, so they're orthogonal.

B.

i. functions with sharp corners/jumps (like a square wave) need infinitely many frequencies, so no finite basis can span the space

ii. compression (a few coefficients instead of a whole table), and the coefficients show which cycles drive the data + you can plug in any t

iii. you can't compute infinitely many terms, so you have to cut it off and its only an approximation

C.

i. $1,\ \cos(2\pi t/12),\ \cos(2\pi t/24)$. the constant level, a 12 hour tide cycle (twice a day), and a 24 hour cycle (once a day).

ii. write $h(t)$ as a linear combination of the basis:

$$h(t) = 2.4\,(1) + 1.8\cos\left(\tfrac{2\pi t}{12}\right) + 0.5\cos\left(\tfrac{2\pi t}{24}\right) \quad \Rightarrow \quad \boxed{[h]_B = (2.4,\ 1.8,\ 0.5)}$$

iii. the average water height (the cosines average to 0)

iv. cosine goes between -1 and 1, so each term can move the water by up to its coefficient:

$$\text{12 hr: } 1.8(\pm 1) = \pm 1.8 \qquad \text{24 hr: } 0.5(\pm 1) = \pm 0.5$$

the 12 hour one, since $1.8 > 0.5$.

D.

i. 440, it has the biggest coefficient (3.5)

ii. drop 0.05:

$$\boxed{s(t) \approx 3.5\sin(440t) + 1.2\sin(880t) + 0.3\sin(1320t)}$$

the energy (loudness) of each frequency is its coefficient squared. since the basis is orthogonal (part A), the total is the sum:

$$3.5^2 + 1.2^2 + 0.3^2 + 0.05^2 = 12.25 + 1.44 + 0.09 + 0.0025 = 13.7825$$

$$\frac{0.0025}{13.7825} \approx 0.00018 \approx 0.02\%$$

so the dropped term is only about 0.02% of the sound, and you cant really hear the difference

Problem 2: Principal Component Analysis

write-up

PCA is a change of basis where the first new axis points where the data spreads out the most. if the data barely moves along the other axes, you can drop them and lose almost nothing. coordinates in $B = \{\mathbf{b}_1, \mathbf{b}_2\}$ come from solving $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2$. we didn't cover this in class, so i loosely learned it from:

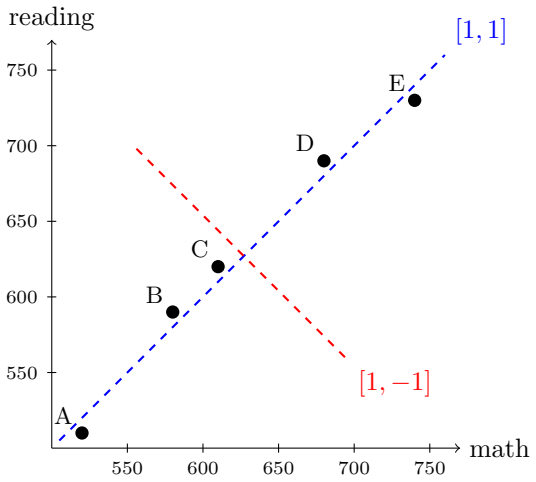
- "StatQuest: Principal Component Analysis (PCA), Step-by-Step" (youtube) `https://www.youtube.com/watch?v=FgakZw6K1QQ`
- ai tools to explain the rest since i was pretty confused.

work

A.

i. (math, reading): $A = (520, 510),\ B = (580, 590),\ C = (610, 620),\ D = (680, 690),\ E = (740, 730)$

ii. & iv.



ii. points almost lie on $y = x$. high math → high reading
iii. no, they spread along $y = x$ but barely perpendicular to it
iv. $[1, 1]$
v.

$$c_1 + c_2 = m, \quad c_1 - c_2 = r \quad \Rightarrow \quad c_1 = \tfrac{m+r}{2}, \quad c_2 = \tfrac{m-r}{2}$$

$$A: \ c_1 = \tfrac{520+510}{2} = \tfrac{1030}{2} = 515, \quad c_2 = \tfrac{520-510}{2} = \tfrac{10}{2} = 5$$
$$B: \ c_1 = \tfrac{580+590}{2} = \tfrac{1170}{2} = 585, \quad c_2 = \tfrac{580-590}{2} = \tfrac{-10}{2} = -5$$
$$C: \ c_1 = \tfrac{610+620}{2} = \tfrac{1230}{2} = 615, \quad c_2 = \tfrac{610-620}{2} = \tfrac{-10}{2} = -5$$
$$D: \ c_1 = \tfrac{680+690}{2} = \tfrac{1370}{2} = 685, \quad c_2 = \tfrac{680-690}{2} = \tfrac{-10}{2} = -5$$
$$E: \ c_1 = \tfrac{740+730}{2} = \tfrac{1470}{2} = 735, \quad c_2 = \tfrac{740-730}{2} = \tfrac{10}{2} = 5$$

$$\boxed{A = (515, 5), \ B = (585, -5), \ C = (615, -5), \ D = (685, -5), \ E = (735, 5)}$$

check A: $515(1, 1) + 5(1, -1) = (515 + 5, \ 515 - 5) = (520, 510)$. ✓
spread along each axis:

$$[1, 1]: \ 735 - 515 = 220 \qquad [1, -1]: \ 5 - (-5) = 10$$

the $[1, -1]$ coordinates are all ± 5, basically nothing compared to the $[1, 1]$ ones
vi. $[1, 1]$ = overall performance (c_1 is the average). $[1, -1]$ = math vs reading difference yes more useful, one number ranks students and the other shows if they are lopsided
vii. $[1, 1]$, you only lose ± 5 points
B.

i. 20 before, 2 after.

ii. less storage, faster, and you can actually plot 2 coordinates. for example with 1000 samples:

$$\text{before: } 1000 \times 20 = 20{,}000 \text{ numbers} \qquad \text{after: } 1000 \times 2 = 2{,}000 \text{ numbers}$$

(plus the 2 basis vectors, $2 \times 20 = 40$ numbers), so about 10 times less.

C. using $[1, 1]$ (total) and $[1, -1]$ (preference): $c_1 = \frac{\text{TV+movies}}{2}$, $c_2 = \frac{\text{TV}-\text{movies}}{2}$

$$A: \ c_1 = \tfrac{2+3}{2} = 2.5, \quad c_2 = \tfrac{2-3}{2} = -0.5$$
$$B: \ c_1 = \tfrac{4+4}{2} = 4, \quad c_2 = \tfrac{4-4}{2} = 0$$
$$C: \ c_1 = \tfrac{6+7}{2} = 6.5, \quad c_2 = \tfrac{6-7}{2} = -0.5$$
$$D: \ c_1 = \tfrac{8+9}{2} = 8.5, \quad c_2 = \tfrac{8-9}{2} = -0.5$$
$$E: \ c_1 = \tfrac{10+11}{2} = 10.5, \quad c_2 = \tfrac{10-11}{2} = -0.5$$

i. total viewing. spread: $10.5 - 2.5 = 8$ vs $0 - (-0.5) = 0.5$

ii. total viewing

iii. TV vs movies. user B stands out ($c_2 = 0$, everyone else -0.5)

iv. yes, it splits "how much" from "what they watch" and nothing is lost (just a change of basis)

Problem 4: Cryptography

write-up

mod 5 means keep the remainder after dividing by 5. since 5 is prime, every number that isnt zero has an inverse, so $\mathbb{Z}_5^2$ works like a normal vector space (just reduce mod 5) encrypting = multiply the message by a secret invertible matrix, decrypting = multiply by the inverse. we didn't cover this in class, so i loosely learned it from:

- this one i was already familiar with modular, but i used ai tools to explain the rest.

work

A. divide by 5 and keep the remainder:

i. $3 + 4 = 7 = 1(5) + 2 \equiv \boxed{2} \pmod 5$

ii. $4 + 4 = 8 = 1(5) + 3 \equiv \boxed{3} \pmod 5$

iii. $3 \cdot 4 = 12 = 2(5) + 2 \equiv \boxed{2} \pmod 5$

iv. try every value:

$$2(0) = 0, \quad 2(1) = 2, \quad 2(2) = 4, \quad 2(3) = 6 = 1(5) + 1 \equiv 1, \quad 2(4) = 8 \equiv 3$$

$$\boxed{x = 3}$$

B. i. 5 choices for the first entry and 5 for the second: $5 \times 5 = \boxed{25}$ vectors

ii. $(2, 4) + (3, 2) = (2 + 3,\ 4 + 2) = (5, 6) \equiv \boxed{(0, 1)}$

iii. $3(1, 4) = (3 \cdot 1,\ 3 \cdot 4) = (3, 12) \equiv \boxed{(3, 2)}$

C.

i. $c_1(1, 0) + c_2(0, 1) = (c_1, c_2) = (0, 0) \Rightarrow c_1 = c_2 = 0$, so independent 2 independent vectors in a 2-dim space, so yes its a basis

ii. $(3, 4) = 3\mathbf{v}_1 + 4\mathbf{v}_2 \Rightarrow \boxed{(3, 4)}$

iii.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & 2 & 4 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right] \Rightarrow \boxed{(2, 1)}$$

check: $2(1, 1) + 1(1, 2) = (2 + 1,\ 2 + 2) = (3, 4)$. ✓

(also $\det \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = 1(2) - 1(1) = 1 \not\equiv 0$, so $\mathbf{u}_1, \mathbf{u}_2$ is a basis)

same vector, different coordinates

iv. if you only see $(2, 1)$ and dont know the basis, you dont know it means $(3, 4)$ the basis is the key

D.

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1(1) + 2(2) \\ 3(1) + 1(2) \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} \equiv \boxed{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1(3) + 2(4) \\ 3(3) + 1(4) \end{bmatrix} = \begin{bmatrix} 11 \\ 13 \end{bmatrix} = \begin{bmatrix} 2(5) + 1 \\ 2(5) + 3 \end{bmatrix} \equiv \boxed{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}$$

A has to be invertible mod 5 ($\det A \not\equiv 0$). but

$$\det A = 1(1) - 2(3) = 1 - 6 = -5 \equiv 0 \pmod 5$$

so this A does NOT work: $(1, 2)$ and $(0, 0)$ both encode to $(0, 0)$, so you cant get $(1, 2)$ back

E.

i. 100 (each one is a byte, 0–255)

ii. 257 is prime, so every nonzero number has an inverse. in $\mathbb{Z}_{256}$ even numbers dont, so matrix inverses can fail. example with 2:

$$\mathbb{Z}_{257}: \ 2(129) = 258 = 1(257) + 1 \equiv 1, \text{ so } 2^{-1} = 129$$

$$\mathbb{Z}_{256}: \ 2x \text{ is always even, and } 2(128) = 256 \equiv 0, \text{ so } 2x \not\equiv 1 \text{ for any } x$$

iii. messages are vectors, the standard basis is the plain message, and the secret basis is the key. encrypt = change to the secret basis, decrypt = change back.

iv. matrices are how you do a change of basis, they mix all the characters together (harder to crack), and decrypting is multiplying by A^{-1}???